

Product Requirements Document

Smart Building Energy Management System

1. Product Summary

The Smart Building Energy Management System helps facilities teams monitor, analyze, and optimize energy usage across commercial buildings. The system collects telemetry from meters, sensors, and building automation systems; identifies abnormal energy behavior; recommends energy-saving actions; and helps operators balance efficiency, cost, and occupant comfort.

The product is intended for office buildings, campuses, managed commercial properties, and multi-building portfolios that want better energy visibility, reduced operating cost, and improved operational response to energy and equipment issues.

2. Product Goals

The product should:

1. Provide real-time visibility into building energy usage.
2. Collect and normalize telemetry from meters, sensors, and building automation systems.
3. Detect abnormal consumption, equipment issues, and stale telemetry.
4. Recommend actions that reduce energy waste while maintaining comfort.
5. Support operator review and approval of optimization actions.
6. Integrate with utility pricing, weather, notification, and building automation systems.
7. Provide reporting for energy performance, estimated savings, and operational issues.
8. Maintain audit history for alerts, recommendations, approvals, overrides, and control actions.

3. Non-Goals

The product will not:

1. Replace the building automation system.
2. Replace low-level equipment controllers or embedded device firmware.
3. Directly manage life-safety systems such as fire alarms, emergency ventilation, access control, or emergency lighting.
4. Guarantee energy savings without proper configuration, sensor coverage, and operational adoption.
5. Provide tenant billing or utility invoice reconciliation in the initial release.

6. Perform structural engineering analysis.
7. Make irreversible or high-risk equipment changes without authorized configuration and approval.

4. Target Users

User	Description
Facilities Manager	Oversees building performance, energy goals, alerts, schedules, and operational policy.
Building Operator	Responds to alerts, reviews recommendations, updates operating schedules, and manages daily building conditions.
Sustainability Analyst	Reviews energy trends, carbon impact, savings estimates, and portfolio performance.
Occupant	May provide comfort feedback or report environmental issues.
System Administrator	Configures buildings, integrations, users, roles, thresholds, and automation policies.

5. External Systems and Devices

External System or Device	Description
Building Automation System	Provides equipment state and executes approved HVAC, lighting, schedule, or setpoint commands.
Energy Meter	Provides electricity, gas, water, or submeter readings.
Environmental Sensor	Provides temperature, humidity, occupancy, CO ₂ , light-level, or other environmental readings.
Utility Pricing Feed	Provides time-of-use pricing, tariff, peak demand, or demand-response event data.
Weather Data Provider	Provides current and forecasted weather data.
Notification Provider	Sends email, SMS, or messaging notifications.
Identity Provider	Optional system used to authenticate internal users and resolve roles or groups.

6. Primary User Problems

Facilities teams often lack a unified view of energy usage across buildings, floors, zones, meters, sensors, and equipment. Energy waste may go unnoticed when HVAC or lighting runs outside occupancy hours, when sensors fail silently, or when schedules no longer match actual building usage.

Operators also need confidence that recommendations and automated adjustments will not create comfort problems or operational risk. The system should make energy performance observable, explainable, and actionable.

7. Product Capabilities

7.1 Telemetry Collection

The system should collect telemetry from building meters, sensors, gateways, and automation systems.

Telemetry may include:

1. Electricity consumption.
2. Gas or thermal energy usage.
3. Water usage where configured.
4. Zone temperature.
5. Humidity.
6. Occupancy.
7. CO₂ level.
8. Lighting level.
9. Equipment operating state.
10. HVAC runtime.
11. Control setpoints.
12. Last device communication timestamp.

The system should normalize telemetry into a consistent internal format so dashboards, alerts, recommendations, and reports can operate across different devices and buildings.

7.2 Real-Time Monitoring Dashboard

Facilities users should be able to:

1. View current energy usage by building, floor, zone, meter, or equipment group.
2. View current environmental conditions.
3. See active alerts.
4. Compare current usage against expected baselines.
5. View telemetry freshness and device connectivity status.
6. Drill into recent trends for selected buildings, zones, meters, or equipment groups.

7. Identify high-consumption areas quickly.

The dashboard should make abnormal conditions visible without requiring users to inspect raw telemetry streams.

7.3 Energy Anomaly Detection

The system should identify abnormal energy and operational conditions.

Detected conditions may include:

1. Energy consumption above expected baseline.
2. Equipment running outside scheduled hours.
3. Sudden demand spikes.
4. Simultaneous heating and cooling.
5. Comfort threshold violations.
6. Missing, stale, or invalid sensor readings.
7. Failed control commands.
8. Integration failures.
9. Unusual runtime patterns.
10. Demand-response underperformance.

Each anomaly should include severity, affected location or equipment, supporting evidence, likely cause where available, and recommended next action.

7.4 Optimization Recommendations

The system should generate recommendations that reduce energy use or cost while respecting comfort and operational constraints.

Recommendations may include:

1. Adjusting HVAC schedules.
2. Modifying temperature setpoints within approved ranges.
3. Reducing lighting during low-occupancy periods.
4. Shifting flexible loads away from high-cost pricing windows.
5. Correcting equipment schedules.
6. Investigating sensors with suspicious readings.
7. Participating in a demand-response event.
8. Returning equipment to normal operation after an optimization period.

Each recommendation should include:

1. Human-readable explanation.
2. Affected building, zone, device, schedule, or equipment group.
3. Estimated impact where available.

4. Confidence level or rationale.
5. Comfort or operating constraints considered.
6. Required approval status.
7. Expected duration if the action is temporary.

7.5 Approved Control Actions

Where enabled, authorized users should be able to approve recommended control actions.

Approved actions may include:

1. Updating HVAC setpoints.
2. Adjusting operating schedules.
3. Enabling energy-saving modes.
4. Applying demand-response adjustments.
5. Restoring normal operating settings.
6. Sending control commands to the building automation system.

The system should enforce configured guardrails before sending a control command. Operators should be able to override automated or approved actions when authorized.

7.6 Utility Pricing and Demand Response

The system should use utility pricing and demand-response data where available.

The system should support:

1. Time-of-use price windows.
2. Peak demand thresholds.
3. Demand-response event notifications.
4. Upcoming high-cost period alerts.
5. Recommendations based on forecasted price changes.
6. Load-reduction planning for eligible buildings and equipment.
7. Post-event performance summaries.

Facilities managers should be able to approve or reject demand-response participation unless automatic participation has been explicitly configured.

7.7 Weather-Aware Optimization

The system should use weather data to improve optimization decisions.

Weather inputs may include:

1. Current outdoor temperature.
2. Forecasted temperature.
3. Humidity.

4. Severe weather indicators.
5. Solar or daylight conditions where available.

The system should avoid recommendations that could create comfort or operational issues under forecasted weather conditions.

7.8 Occupant Comfort Feedback

Where enabled, occupants should be able to submit comfort feedback or environmental issue reports.

Feedback may include:

1. Too hot.
2. Too cold.
3. Poor air quality.
4. Lighting issue.
5. Noise or equipment issue.
6. Free-form comment.
7. Location or zone.

The system should correlate comfort feedback with telemetry and recent optimization actions where possible.

7.9 Reporting and Analytics

The system should provide reporting for:

1. Energy usage trends.
2. Energy cost trends.
3. Baseline versus actual performance.
4. Optimization recommendations and actions.
5. Estimated savings.
6. Alert history.
7. Equipment runtime.
8. Demand-response participation.
9. Telemetry freshness.
10. Carbon impact estimates where emissions factors are configured.
11. Building and portfolio benchmarking.

Reports should support filtering by building, zone, equipment group, date range, and metric type.

8. Representative User Journeys

8.1 Monitor Building Energy Performance

1. Facilities Manager opens the monitoring dashboard.
2. System displays current energy usage, environmental status, and active alerts.
3. Facilities Manager selects a building with elevated usage.
4. System shows usage by floor, zone, meter, or equipment group.
5. Facilities Manager reviews related alerts and recommendations.
6. Facilities Manager assigns or performs follow-up action.

8.2 Respond to Energy Anomaly

1. System detects HVAC energy usage above expected baseline.
2. System creates an alert with severity and supporting evidence.
3. Building Operator receives a notification.
4. Building Operator opens the alert.
5. Building Operator reviews affected zone, telemetry, and equipment state.
6. Building Operator acknowledges the alert.
7. Building Operator applies a recommended correction or dismisses the alert with a reason.
8. System records the action.

8.3 Apply Optimization Recommendation

1. System identifies that a floor is unoccupied after hours while HVAC remains active.
2. System creates an optimization recommendation.
3. Facilities Manager reviews expected impact and comfort constraints.
4. Facilities Manager approves the recommendation.
5. System sends the approved schedule adjustment to the building automation system.
6. Building automation system acknowledges or rejects the command.
7. System records the result.
8. System monitors post-action telemetry.

8.4 Participate in Demand-Response Event

1. Utility pricing feed provides a demand-response event.
2. System evaluates eligible buildings and controllable loads.
3. System generates proposed load-reduction actions.
4. Facilities Manager approves participation.
5. System sends approved commands to the building automation system.
6. System monitors energy reduction during the event.
7. System restores normal operating settings after the event.
8. System generates an event performance summary.

8.5 Review Sustainability Performance

1. Sustainability Analyst opens the reporting view.
2. Analyst selects a portfolio, building, and date range.
3. System displays usage, cost, savings, and carbon impact estimates.
4. Analyst compares actual performance against baseline.
5. Analyst exports or shares the report with stakeholders.

9. Functional Requirements

ID	Requirement
FR-001	The system shall collect telemetry from energy meters.
FR-002	The system shall collect environmental telemetry from building sensors.
FR-003	The system shall collect equipment and control state from building automation systems where configured.
FR-004	The system shall normalize telemetry into a common internal format.
FR-005	The system shall display real-time energy and environmental status.
FR-006	The system shall identify stale, missing, or invalid telemetry.
FR-007	The system shall detect abnormal energy consumption.
FR-008	The system shall generate alerts for energy, comfort, telemetry, equipment, and integration conditions.
FR-009	The system shall notify users about high-severity alerts.
FR-010	The system shall generate energy optimization recommendations.
FR-011	The system shall explain the rationale for each recommendation.
FR-012	The system shall allow authorized users to approve or reject recommendations.
FR-013	The system shall send approved control commands to the building automation system.
FR-014	The system shall record control command

ID	Requirement
	status and result.
FR-015	The system shall allow authorized users to override automated or approved actions.
FR-016	The system shall ingest utility pricing or demand-response data where configured.
FR-017	The system shall ingest weather data where configured.
FR-018	The system shall provide energy usage reports by building, zone, equipment group, and time range.
FR-019	The system shall maintain an audit trail of alerts, recommendations, approvals, overrides, and control actions.
FR-020	The system shall support configuration of buildings, floors, zones, meters, sensors, equipment groups, operating schedules, and comfort thresholds.
FR-021	The system shall support occupant comfort feedback where enabled.
FR-022	The system shall expose failed integrations and command failures to authorized operational users.

10. Non-Functional Requirements

Category	Requirement
Reliability	The system shall tolerate temporary device, network, and external integration failures without losing critical operational state.
Availability	Monitoring and alerting capabilities should remain available during normal building operating hours.
Data Freshness	The system shall expose telemetry freshness so users can distinguish current readings from stale data.
Safety	Automated and approved control actions shall be constrained by configured comfort, safety, and operational limits.
Auditability	The system shall record who or what initiated recommendations, approvals,

Category	Requirement
	overrides, and control commands.
Performance	Real-time dashboards should load recent building status quickly under normal portfolio usage.
Scalability	The system should support growth from a single building to a multi-building portfolio.
Security	User access shall be authenticated and role-based. Control actions shall be restricted to authorized users or approved automation policies.
Explainability	Recommendations and automated actions shall include enough rationale for facilities users to understand why they were generated.
Interoperability	The system should support integration with common building automation, sensor, meter, pricing, weather, and notification sources through adapters or standard protocols where feasible.
Resilience	Telemetry ingestion should buffer or recover data during temporary downstream processing failures.
Privacy	Occupancy and comfort feedback data shall avoid unnecessary personal identification.
Observability	Administrators shall be able to identify failed integrations, delayed telemetry, alert backlogs, and command failures.

11. Data Requirements

The system should manage the following data:

1. Building portfolio structure.
2. Floors, zones, and equipment groups.
3. Meter registry.
4. Sensor registry.
5. Device-to-zone mappings.
6. Equipment registry.
7. Telemetry readings.
8. Normalized time-series data.

9. Operating schedules.
10. Comfort thresholds.
11. Energy baselines.
12. Optimization rules.
13. Optimization recommendations.
14. Recommendation approvals and rejections.
15. Control command history.
16. Alert records.
17. Utility pricing and demand-response events.
18. Weather observations and forecasts.
19. Occupant comfort feedback.
20. Audit events.
21. User roles and permissions.
22. Reporting aggregates.

Historical telemetry should be retained long enough to support trend analysis, baseline calculation, anomaly detection, savings estimation, and reporting.

12. Reporting and Operational Visibility

Facilities users and analysts should be able to view:

1. Current energy usage.
2. Current environmental status.
3. Active alerts.
4. Telemetry freshness.
5. Device connectivity status.
6. Recommendation history.
7. Control command history.
8. Demand-response event performance.
9. Energy usage by building, zone, meter, or equipment group.
10. Cost trends.
11. Estimated savings.
12. Carbon impact estimates where configured.
13. Alert frequency and resolution status.
14. Equipment runtime trends.

13. Success Metrics

Metric	Target Direction
Energy consumption per building operating hour	Decrease

Metric	Target Direction
Peak demand charges	Decrease
After-hours equipment runtime	Decrease
Energy anomalies detected before manual discovery	Increase
Mean time to acknowledge critical facilities alerts	Decrease
Approved optimization recommendations	Increase
Control command failure rate	Decrease
Telemetry freshness across active sensors	Increase
Occupant comfort complaints after optimization actions	Avoid increase
Estimated savings from optimization actions	Increase
Demand-response event participation success	Increase

14. Risks and Open Questions

Area	Question or Risk
Building automation integration	Which automation systems and protocols must be supported first?
Control authority	Which actions may be automated, and which require human approval?
Safety guardrails	What comfort, safety, and equipment constraints must always be enforced?
Sensor reliability	How should the system handle missing, stale, or contradictory sensor data?
Comfort constraints	Who defines acceptable comfort ranges for each building or zone?
Baseline accuracy	How much historical data is required before anomaly detection and savings estimates are reliable?
Tenant impact	Are there tenant-specific comfort, operating schedule, or reporting requirements?
Demand response	Can the system participate automatically, or must each event be approved manually?
Cybersecurity	What controls are required before allowing remote control of building equipment?

Area	Question or Risk
Data granularity	What telemetry resolution is required for monitoring, optimization, and reporting?
Reporting	Which reports are required for facilities operations versus sustainability stakeholders?
Occupant feedback	Should occupant feedback be anonymous, authenticated, or linked to a location only?

15. Release Scope

MVP

The MVP should include:

1. Building, zone, meter, and sensor configuration.
2. Telemetry ingestion from selected meters and sensors.
3. Real-time monitoring dashboard.
4. Telemetry freshness indicators.
5. Basic energy anomaly alerts.
6. Email notifications for critical alerts.
7. Historical energy usage reporting.
8. Manual recommendation review.
9. Audit logging for alerts, recommendations, and user actions.
10. Basic role-based access.

Post-MVP

Future releases may include:

1. Automated HVAC and lighting control actions.
2. Demand-response event support.
3. Weather-aware optimization.
4. Utility price-aware optimization.
5. Occupant comfort feedback.
6. Advanced anomaly detection.
7. Predictive equipment runtime analysis.
8. Carbon impact reporting.
9. Multi-building portfolio benchmarking.
10. Configurable optimization policies.
11. External identity provider integration.
12. Mobile operator experience.
13. Building automation command integration.

14. Advanced operational health monitoring.